

Problem Set 8: NP-completeness II

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Due: 04/13/2023 11:59pm

- Please **TYPE** your solutions using Latex or any other software. Handwritten solutions *won't* be accepted.

Steps to write a reduction

To show that a problem is NP-complete, you need to show that the problem is both in NP and NP-hard. In a polynomial-time reduction, we have a *Problem B*, and we want to show that it is NP-complete. These are the steps:

- Demonstrate that *problem B* is in the class NP. This is a description of a procedure that verifies a candidate's solution. It needs to address the runtime. You should give a polytime verifier which takes as input a candidate problem, and a witness, and outputs true or false if the witness is a solution.
- Demonstrate that *problem B* is at least as hard as a problem previously proved to be NP-complete. Choose some NP-complete *A* and prove $A \leq_p B$. You need to show that *problem B* is NP-hard. This is done via polytime reduction from a known NP-complete *problem A* to the unknown *problem B* ($A \rightarrow B$) as follows:
 1. Choose *A*. You will have many NP-complete problems to pick from later on, and it is best to pick a similar one.
 2. Give your reduction f and argue that it runs in polynomial time.
 3. Prove that $x \in A \implies f(x) \in B$
 4. Prove that $x \notin A \implies f(x) \notin B$
 5. This is sufficient to show $x \in A \iff f(x) \in B$. Since *A* was NP-complete, and $A \leq_p B$, we can conclude that *B* is NP-hard. Note: You don't have to provide a formal proof. You can briefly explain in words both implications and why they hold.

The second bullet point above is prove that *problem B* is NP-hard which combined with the first bullet point yields the NP-complete proof.

Problem 1

(30 points)

Show that the following **Party-Invitation** problem is NP-Complete.

Input: You are given a set of guests G of length m to be invited to a party, and a set of k individual family sets $F_1, F_2, \dots, F_k$ such that they are non-disjoint and $F_i \subseteq G$.

Output: A set Q of guests such that not all members of any one family F_i are invited to the party.

Note: Write the reduction in the format described on the front page of this document.

has to output at least n guests

Think of reducing this from some already NP-complete graph problem and see how they correlate.

Here, non-disjoint signifies that two sets might have the same family member A . It's not necessary that they are strictly non-disjoint. In some cases, you can have all the sets as disjoint. I see where the confusion is coming from.

You can assume they can be disjoint as well. But if the problem works under the non-disjoint case, then the disjoint case is fairly simple.

Problem 2

(30 points)

definitely go from subset sum

Show that the following **Average-Sum** problem is NP-Complete.

Input: a set S of natural numbers.

Output: A subset $S' \subset S$ such that the sum of the elements in the subset S' equals the average of elements in the set S i.e.

$$\sum_{s' \in S'} s' = \frac{1}{|S|} \sum_{s \in S} s$$

Note: Write the reduction in the format described on the front page of this document.

not sure what we add to the set, but it needs to 0 and a number
greater than t (the largest number in the set?)

so if we can reach a target t, we need to add numbers to the set
so that that target is now the mean

if we cannot reach t, then after the adjustment, it is now the mean
and we still cannot reach it

To prove that Average-Sum is NP-Complete, we need to prove it is both in NP and NP-Hard. Let's start with NP. The witness is a subset, S' , of S . To prove that it is in NP, we need to be able to verify the solution in polytime. In order to check if the given subset adds up to the mean of the elements, we need to first calculate the mean which involves looping over the total set S and adding them together and dividing by the length of S . This would run in $O(|S|)$. Then we add the numbers in the subset S' and see if it equals the calculated mean. This part would run in $O(|S'|)$. So the overall verification runtime is $O(|S| + |S'|)$ which is polytime.

Problem 3

(30 points)

def go from indset (or clique)

Show that the following CLIQUE-&-INDEPENDENT-SET problem is NP-Complete.

Input: an undirected graph $G = (V, E)$ and an integer k .

Output: Returns both a k -clique and k -independent set in G i.e. a set $Q \subseteq V$ of k vertices that are fully connected, AND another set $R \subseteq V$ of k vertices with no edges between them. Sets R and Q might not be disjoint.

Note: Write the reduction in the format described on the front page of this document.

so go from indset or clique and basically add the other one

if ind but not a clique then false and vice versa, only true if both are true

indset means nothing in the set are touching
clique means all pairs are touching

start from indset
duplicate the graph, run the given reduction to get to clique
then you have the two sets

Problem 4

(30 points)

Show that the following CYCLE-FREE problem is NP-Complete.

Input: a directed graph $G = (V, E)$ and an integer k .

Output: Returns a set $M \subseteq E$ of size at most k such that $|M| \leq k$ and a subgraph $G' = (V, E \setminus M)$ is cycle-free.

Note: Write the reduction in the format described on the front page of this document.

literally no idea

Problem 5 (*Optional, but highly recommended for the exam*)

Note: We won't grade this problem, but it's highly recommended that you solve the problem yourself.

Show that the following **4-coloring** problem is NP-Complete.

Input: An undirected graph $G = (V, E)$ and an integer $k = 4$.

Output: A 4-coloring of G if exists, else output NO